University of St.Gallen

Course and Examination Fact Sheet: Spring Semester 2026

8,859: Marketing Analytics & Behavioural Insights

ECTS credits: 4

Overview examination/s

(binding regulations see below)
decentral - Oral examination and technical discussions, Digital, Group work group grade (50%)
Examination time: Term time
decentral - Written examination, Digital, Individual work individual grade (50%, 90 mins.)
Examination time: Term time

Attached courses

Timetable -- Language -- Lecturer
8,859,1.00 Marketing Analytics & Behavioural Insights -- English -- Hildebrand Christian Alexander

Course information

Prerequisites

The course does not have hard requirements but the following background can be helpful:

- Basic statistics knowledge is helpful but not mandatory (descriptive and inferential statistics).
- Working knowledge of R (an introduction to R will be provided as part of the course).

Learning objectives

- Learn to apply your technical expertise to better predict market and consumer outcomes to drive innovation in product development and user experience design.
- Employ multivariate techniques to uncover optimal brand positioning strategies based on customer perceptions, preferences, and market data.
- Leverage analytical modeling approaches to reveal and target core customer segments with common characteristics and behaviors.
- Identify, quantify, and optimize key performance indicators to make more effective business and marketing decisions.
- Translate analytical insights into clear strategic recommendations for branding, positioning, segmentation, targeting, and marketing mix decisions.
- Apply state-of-the-art marketing analytics tools to solve real-world problems and make better, data-driven decisions.

Content

The majority of innovations fail because there is a disconnect between what companies develop and what customers actually need or want. This course bridges that gap by offering you a set of data-driven tools to deeply understand customer needs, identify untapped market opportunities, and create differentiated products and services that people value. You will learn foundational marketing analytics techniques to determine which features customers prefer, choice modeling to simulate demand for new product or service offerings, and advanced segmentation approaches to reveal underserved market segments or identify profitable niches in current or future markets. Whether working on breakthrough innovations or improving existing products and services, these methods allow you to engineer experiences that are designed for what consumers are looking for vs. what is technically possible. My hope is that by the end of this course, you expanded your business and marketing knowledge, your methodological toolbox, and to solve real-world problems to develop better products and services that create value in the long-term.

Structure and indications of the learning and teaching design

Session 1: Bridging Computer Science and Marketing Analytics

1. Introducing the integral role of modern marketing analytics and learn why many technical innovations fail due to a lack of addressing a real market need.
2. Learn how marketing analytics offers methodological approaches to reduce such risks and support more effective product development, targeting the right customers, and ultimately drive more effective innovation.

Session 2: Market & Customer Segmentation (Method: Clustering)

1. **Clustering Fundamentals**: Exploring various clustering techniques such as K-means, hierarchical, and DBSCAN, and understanding how to apply them to market and customer data for segmentation.
2. **Segment Profiling and Interpretation**: Techniques for profiling clusters to understand the defining characteristics of different market segments and how to interpret the results for actionable marketing insights.
3. **Segmentation Strategy**: Developing strategies for targeting and positioning based on the identified segments, including product differentiation and personalized marketing.

Session 3: Brand & Product Positioning (Method: Multidimensional Scaling - MDS)

1. **Theory and Application of MDS**: Understanding the basics of multidimensional scaling and its application in visualizing the positioning of brands and products in a multi-dimensional space based on consumer perceptions and preferences.
2. **Interpreting MDS Output**: Learning how to interpret the spatial maps generated by MDS, including dimensions, distances, and the relative positioning of brands and products.
3. **Strategic Implications**: Discussing how to use insights from MDS to inform branding strategy, product development, and competitive positioning.

Session 4: Driver Analysis (Method: Linear Regression)

1. **Linear Regression in Marketing**: Understanding how linear regression can be used to identify and quantify the drivers of consumer behavior and marketing outcomes.
2. **Modeling Marketing Mix Effectiveness**: Developing linear regression models to test hypotheses about relationships between variables, such as the impact of marketing mix elements on sales.
3. **From Drivers to Decisions**: Translating regression analysis into marketing decisions, including identifying key drivers of customer satisfaction and loyalty, and informing strategic decisions in practice.

Session 5: Targeting (Method: Logit Regression)

1. **Logit Models in Marketing**: Learn how to use logistic regression models to predict the probability of critical marketing events (e.g., targeting decisions, purchasing outcomes, customers-at-risk decisions).
2. **Model Building and Validation**: Covering the steps in building a logit regression model, including variable selection, model fitting, and validation techniques.
3. **Applications in Targeting**: Applying logit regression to develop targeting strategies, such as identifying potential customers who are most likely to respond to a particular marketing action.

Session 6: Product Profiling (Method: Conjoint Analysis)

1. **Conjoint Analysis**: Introducing conjoint analysis as a technique to measure consumer preferences regarding various attributes of a product or service.
2. **Designing Conjoint Studies**: Learning how to design conjoint experiments, including selecting attributes, levels, and creating survey designs that effectively capture preference data.
3. **Interpreting and Applying Conjoint Results**: Analyzing conjoint data to determine the relative importance of different attributes and how to use this information for product profiling, feature optimization, and pricing strategies.

Session 7: Market Simulation (Group Case Analysis)

1. **Market Simulation**: Students will engage in a half-day, interactive CRM (customer relationship management) simulation to analyze performance metrics of a firm and experiment with various strategies for customer acquisition and retention. This is a competitive simulation in which all students play against each other.

2. **Reflection and Integration of Learning**: Groups have to make multiple decisions corresponding to key concepts learned in class (e.g., product and brand positioning; product profile investments; targeting new customer segments, etc). This simulation integrates the key concepts interactively, in a competitive but controlled, risk-free environment.

Literature

Mandatory Readings:

- Palmatier, Robert W., J. Andrew Petersen, and Frank Germann. *Marketing Analytics: Based on First Principles*. Bloomsbury Publishing, 2022.
- Beckwith, Harry. Selling the invisible: A field guide to modern marketing. Grand Central Publishing, 1999.

Optional Readings:

- Hildebrand, C. (2019): The Machine Age of Marketing, *Marketing Intelligence Review*, Vol. 11(2), p. 10-17.
- Kotler, P., & Levy, S. J. (1969). Broadening the concept of marketing. *Journal of Marketing*, 33(1), 10-15.
- Wedel, M., & Kannan, P. K. (2016). Marketing analytics for data-rich environments. *Journal of Marketing*, 80(6), 97-121.

Additional information

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Examination information

Examination sub part/s

1. Examination sub part (1/2)

Examination modalities

Examination type	Oral examination and technical discussions
Responsible for organisation	decentral
Examination form	Oral examination
Examination mode	Digital
Time of examination	Term time
Examination execution	Asynchronous
Examination location	Off Campus
Grading type	Group work group grade
Weighting	50%
Duration	--

Languages

Question language: English
Answer language: English

Remark

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Examination-aid rule

Open Book

Students are free to choose aids, apart from the following restrictions:

- pocket calculator models which are not part of the Texas Instruments TI-30 series, as well as any programmable electronic devices that are capable of communication such as electronic dictionaries, notebooks, tablets, smartphones, headsets, additional screens, etc. are not admissible;

- there is an option for faculty members to explicitly define exceptions under supplementary aids.

Procuring any aids, as well as ensuring their working order, is the exclusive responsibility of students.

Supplementary aids

This is an open-book examination and graded based on the performance of the market simulation.

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2. Examination sub part (2/2)

Examination modalities

Examination type	Written examination
Responsible for organisation	decentral
Examination form	Written exam
Examination mode	Digital
Time of examination	Term time
Examination execution	Synchronous
Examination location	On Campus
Grading type	Individual work individual grade
Weighting	50%
Duration	90 mins.

Languages

Question language: English
Answer language: English

Remark

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Examination-aid rule

Closed Book

The use of aids is prohibited as a matter of principle, with the exception of pocket calculator models of the Texas Instruments TI-30 series and, in case of non-language exams, bilingual dictionaries without any handwritten notes. Any other aids that are admissible must be explicitly listed by faculty members in the paragraph entitled “Supplementary aids” of the course and examination fact sheet; this list is exhaustive.

Procuring any aids, as well as ensuring their working order, is the exclusive responsibility of students.

Supplementary aids

This is a closed book exam.

Content

Written Exam (Individual Examination; 50%)

In the written exam, students will test their knowledge about the substantive and methodological topics they have learned about in the first part of the course (e.g., positioning, segmentation, targeting; and the methodological foundations such as multi-dimensional scaling, clustering, conjoint product profiling).

Market Simulation (Group Examination; 50%)

Students will engage an a half-day, interactive CRM (customer relationship management) simulation to analyze performance metrics of a firm and experiment with various strategies for customer acquisition and retention. This is a competitive simulation in which all students play in groups against each other.

Literature

In addition to the lecture slides and the content of all in-class discussions, relevant journal articles, digital content and readings will be announced and made available during the course.

Please note

Please note that only this fact sheet and the examination schedule published at the time of bidding are binding and takes precedence over other information, such as information on StudyNet (Canvas), on lecturers' websites and information in lectures etc.
Any references and links to third-party content within the fact sheet are only of a supplementary, informative nature and lie outside the area of responsibility of the University of St.Gallen.

Documents and materials are only relevant for central examinations if they are available by the end of the lecture period (CW21) at the latest. In the case of centrally organised mid-term examinations, the documents and materials up to CW 13 (Monday, 23rd March 2026) are relevant for testing.

Binding nature of the fact sheets:

- Course information as well as examination date (organised centrally/decentrally) and form of examination: from bidding start in CW 04 (Thursday, 22nd January 2026);
- Examination information (supplementary aids, examination contents, examination literature) for decentralised examinations: in CW 12 (Monday, 16 March 2026);
- Examination information (supplementary aids, examination contents, examination literature) for centrally organised mid-term examinations: in CW 14 (Monday, 30 March 2026);
- Examination information (regulations on aids, examination contents, examination literature) for centrally organised examinations: two weeks before ending with de-registration period in CW 15 (Monday, 06 April 2026).